

Assignment 4

E-Commerce site



GOALS

This fourth assignment is meant to give you hands on experience toward the development of a small e-commerce site and to put in application different concepts seen during the *Module 5 - System Design*, *Module 6 - Verbal Communication* as well as *Module 7 - Usability Heuristics*.

You can choose any e-commerce that you wish, either selling clothing, or furniture, or anything that can be sold!! There are technical requirements for your e-commerce site though, so make sure that what you select will allow you to meet these requirements.

By carrying out this assignment, you will:

- Reflect on the notion of System Image and the compatibility of knowledge organization between the system and the user
- Explore visual design in relation to appropriate layouts for different interactive processes
- Learn how to implement a faceted search
- Reflect on language in relation to various communicative goals
- Take into consideration all usability heuristics in your design and implementation
- Continue expanding your JavaScript development skills
- Learn hands on experience doing a heuristic evaluation of your own design and of others

SEG3125 – Analysis and Design of User Interfaces



SCHEDULE / DEADLINES

Here are the important calendar events for this assignment.

- **Mondays, June 22nd and June 29th tutorials**

Make sure to attend both June 22 and 29 tutorials to better understand the requirements for this assignment, understand how to do a faceted search and perform a heuristic evaluation.

- **Forum monitoring on Brightspace between Tuesday June 23rd and Friday July 3rd**

If something is not clear, if you need help about the assignment, then please ask questions on Brightspace forum. Make sure you get help. Any question is fine, as long as it is asked in a polite and respectful manner. TAs will be monitoring the forum regularly and answer your questions. Not only will their answers help you, they will help other students in the class.

- **Emails to TAs between Tuesday June 23rd and Friday July 3rd (if needed)**

ONLY if the forum help is not sufficient, you can request a one-on-one help session from the TA assigned to your lab session (make sure you contact the TA for YOUR session – Z01 to Z04).

- **Report submission: Sunday, July 5, midnight**

You MUST submit your report, which will follow the requirements and contain a link to your portfolio by Sunday, July 5th, midnight. *No extension will be given. The only exceptions will be for justified medical reasons with medical certificates. The penalty is 20% per day. There is a grace period till 1am... but then it's -20%!*

- **Group review registration: Sunday, July 5, midnight**

Make sure you put your name on the group review schedule. You MUST participate in the group review to practice your analysis skills and get feedback on your design.

- **Group review participation between July 9th and July 15th**

Your scheduled group review session will be within 10 days of your submission. Not registering or missing the group review is 20% off your assignment.



ONLINE TUTORIALS / TECHNOLOGIES

Resources for JavaScript development have been provided in the requirement documents for Assignment 1 and 2.

During the June 22nd tutorial, ***the TA will provide examples for the development of faceted search*** and during the June 29th tutorial, ***the TA will go through a heuristic evaluation.***



REQUIREMENTS

1. Explore e-commerce sites and choose the domain of your site

Spend a bit of time exploring e-commerce sites (e.g. electronics, clothing, groceries, furniture, etc).

Find at least two resources that are inspiring you to develop your site. Decide on what you wish to be selling, to whom, and give a name to your e-commerce.

2. Reflect / Design

Spend time thinking about the various elements that need to be included in your e-commerce site. For this assignment, you are not required to create personas and storyboards. Yet, I do STRONGLY encourage you do mockups even if you will not be required to submit them. In your report, you will be required to describe your design choices, and if you can support those with mockups, that will make your explanations easier.

(A) System Design

Your site will need to include three interactive processes:

- a. A “follow instructions” process for a user to buy an item, with steps such as: put in the cart, enter personal information, pay, get confirmation.
- b. A “exploring in the divergent/convergent process”, using a faceted search. Think of the various features that a user would like to be able to search on. What would be in their semantic network? For example, for a clothing stores, users might be interested in size, color, material, etc. You will not be required to submit the semantic network, but I encourage you to build one to better develop your faceted search.
- c. A “communicate process”, with as a survey to be performed after visiting (or buying).

(B) Verbal Communication

Your site will need to use language (words, sentences) in a thoughtful way.

- d. Think of your Writer/Reader Model. Decide on how the site will present itself and will be talking to the users.
- e. Think of the intention/purpose conveyed by your sentences. You are required to include at least three types of communication purpose in your site:
 - Incite to action: Use promotional language somewhere on your site showing a user some “deals” or rebates and try to make the user buy items.
 - Inform: Use an informative tone to allow the user to perform their shopping and understand the characteristics of the different items.
 - Engage in a connection: Use a language that is positive and engaging for encouraging the users to fill the survey.

3. Implement your e-commerce site as a high-fidelity prototype

Using a *JavaScript framework* (e.g., *React*), create a high-fidelity interactive.

The high-fidelity prototype should meet the following technical requirements:

- a. Be developed using a JavaScript framework (e.g., *React*)
- b. Make use of adequate organization, input and navigation patterns (e.g. patterns facilitating the browsing through and buying of an item)
- c. Be interactive (allow a user to actually browse, buy an item, fill-in the survey).
 - Product exploration must be functional with a faceted search.
 - Payment does not need to be linked to any payment platform. You just need to include a form for users to include payment information.
 - The survey can be simple, just asking users to comment/evaluate their experience on the site.
- d. Be accessible through your portfolio (from assignment 1)

The high-fidelity prototype should be in line with the reflection done in step 2 above:

(A) Interactive Processes and System Image

- The 3 required interactive processes should be clearly found on the site.
- The “follow instructions process” should clearly show what step the user is at and what is left to do and what has been done.
- The “communicate process” does need to be realized with a short survey.
- The “divergent/convergent explore process” must be implement with a faceted search that is well thought out and functional.

(B) Verbal Communication

- Your choice of Writer/Reader model should be clear and consistent. For “conversation effects” (see for example the use of “I” in the case study on conversation in the video lecture) the model can be locally changed in a few places.
- The 3 communicative purposes (incite to action, inform, engage in a connection) should be recognizable from the use of proper words and types of sentences (e.g. exclamatory, interrogative).

Your high-fidelity prototype should be guided by visual design principles:

- a. Use a color theme and be consistent
- b. Use typography to attract attention
- c. Think of the impact of the layout and the use of negative space
- d. Use well the principles of contrast, scale, balance, visual hierarchy
- e. Think of the Gestalt Principles for Perception (e.g. similarity, continuity, closure, etc).
- f. Find a good synergy between visual communication and verbal communication so that both work together.

4. Perform a heuristic evaluation of your site

In Module 7, you are learning about 10 usability heuristics. Those were developed for predictive evaluation to foresee difficulties and issues that might arise if the heuristics are not followed. Therefore, even if it's presented as an evaluation, I suggest that you use the heuristics to further guide your design and implementation. I could have made that suggestion in Step 2 and 3 above, but I prefer to talk about the heuristics in a separate section so you think of the heuristics as both design-guiding and design-evaluation tools.

As a requirement for this assignment you must go through all 10 usability heuristics and for each one, point to an element in your design that you find particularly in line with the heuristic. You can include a small screenshot of that element to make it easy to see. For all 10 heuristics, make sure you provide the name of the heuristic (do not refer to it by a number) and an example from your site.

5. Link your portfolio to your high-fidelity prototype

You created your portfolio during Assignment 1. You must now link your portfolio to the high-fidelity prototype of the e-commerce as you did for the service site (Assignment 2) and the memory game (Assignment 3). You can also (if you wish) link your portfolio to your two storyboards to demonstrate your working method and contribute to the expertise (*how you work*) section of your portfolio.

Your portfolio and memory game should be accessible to the public (or, if you prefer, password protected with the password provided in your report).

The TA must be able to test your UI without having to communicate with you. This is ESSENTIAL.

6. Write a short report describing your design process, choices and heuristic evaluation:

Your report should include:

1. Designer: Name, Student Number
2. E-Commerce Goal:
 - a. What is the name and type of e-commerce you decided to build.
 - b. Links to the inspiration sites with a description of how they were used for inspiration.
3. Reflection/Design:
 - (A) Interactive process / System Image design:
 - a. For the follow instruction process, explain what you decided to use for providing the user with feedback as to what step they are at and what remains to be done?
 - b. For the explore process, describe the filters you decided to include in your faceted search.
 - c. For the communicate process, describe how the survey could be not too intrusive and engaging.
 - (B) Verbal communication design:
 - Explain what you decided the Writer/Reader model (“who talks to who”) and how you insured consistency with perhaps some variations for “conversation effects”.
 - Give examples of choices of words and sentence types that you used for the 3 communicative purposes (incite to action, inform, engage in a connection).
4. High-fidelity prototype
 - a. Explain the visual design choices you made.
 - b. Link to your portfolios – when working in pairs, provide links to both portfolios. Your prototype should be accessible through each portfolio.
5. Code: A link to the group’s GitHub
6. Heuristic Evaluation: As described in Step 4 of the requirements section, you must demonstrate how your site is in line with each of the 10 heuristics.
7. Generative AI Acknowledgement: Many of you will use generative AI to help you create your high-fidelity prototypes and your report. Describe the role of generative AI for each of those, providing the GenAI tools used and your interactions with them.



EVALUATION

This assignment is worth 10% of your semester and is evaluated on 100 points. The 10% weight reflects the fact that you are performing a heuristic evaluation on top of doing your design and implementation.

- Your report's content and clarity (5 points)
 - Report easy to read and answers the required questions.
 - Report contains all the required elements with answers to better understand your design choices and the help provided by GenAI.
 - Different sections of the report are well identified
 - Information is provided in a concise manner, avoiding long paragraphs.
- Your reflection and design choices for System Image / Interactive Processes (5 points)
- Your reflection and design choices for Verbal Communication (5 points)
- Your hosted web site, accessible, made according to the requirements provided (50 points)
 - UI should reflect the choices made in accordance with the usability heuristics
 - UI should reflect design choices for interactive processes
 - UI should reflect design choices for verbal communication
- Your active participation in the group review (20 points)
 - Ability to analyze other student's design according to the principles explored in the lecture
 - Ability to answer questions from TA and other students
- Your self heuristic evaluation (15 points)
 - Heuristic evaluation should clearly present each heuristic and point (and describe) to an element in the design that is in accordance with the heuristic.



SUBMISSION CONTENT / PLATFORM

Each student must submit their report, PDF version (see all its requirements above) on Brightspace. The link can be found in the Assignment section, for Assignment 4.

ALSO, very important, **MAKE SURE you register on the signup sheet for your group review.** You will be informed when the signup sheet is available.



QUESTIONS

- See the section Schedule/Deadline of the current document. It indicates tutorials and forum for help and questions.
- Please DO NOT send the professor any emails. The forum help should be enough to obtain answers to your questions. Do not be shy to ask questions on the forum. Any answers provided there by the TA will help many other students.
- If the forum help is not sufficient, email the TA assigned to your lab session (Z01 to Z04)